

Katherine M. Yut

Geospatial Data Engineer

Expert in geospatial analysis, statistics, and programming using R and Python. Passionate about data visualization. Seeking to solve interesting environmental problems. Meticulous, intrinsically motivated, and a quick learner.

Work History

Grassroots Carbon, San Antonio, TX (hybrid)

Data Engineer 2025-08 – current

Building ETL/ELT data pipelines using Python, GitHub, and Docker containerization.

Skills: pydantic, Streamlit, cron, GitHub Actions, CI/CD.

Grassroots Carbon, San Antonio, TX (hybrid)

Carbon Technical Manager 2024-08 – 2025-08

Automated rancher eligibility screening processes using R shiny and Python, provided geospatial data analysis using QGIS, and supported the navigation of carbon standards.

University of Texas, Bureau of Economic Geology, Austin, TX

GIS & Energy Data Analyst 2022-05 – 2024-07

Leveraged R to clean and manipulate data tables, solve problems, and automate processes; utilized ArcMap to aid in oil & gas research publications; and published research findings as papers, posters, and presentations.

Education

Master of Arts: Economics and Big Data

Bachelor of Arts: Economics and Big Data

Bachelor of Science: Geographic Information Science

University of Oklahoma (Norman, OK)

2016-08 – 2021-05, *Summa cum laude*

Research

KM Smye, **K Yut**, RC Reedy, BR Scanlon, JP Nicot, P Hennings (2024). Challenges with managing unconventional water production and disposal in the Permian Basin. AAPG Bulletin 108 (12), 2215-2240.

Peng, S., Maraggi, LMR., Bhattacharya, S., **Yut, K.**, McMahon, T., Haddad, M. (2023). Feasibility of CO2 storage in depleted unconventional oil and gas reservoirs: Capacity, microscale mechanism, injectivity, fault stability, and monitoring. URTEC 13–15 June 2023.

M Bull, T Francis, **K Yut**, K Weber, J Spruce, C Jarnevich. West Virginia Ecological Forecasting. NASA DEVELOP National Program, 2019.

Kelley, S., **Yut, K.**, Kulkarni, R., & Gaffin, D. (2019). Avoidance of rosemary oil by scorpions. Journal of Arachnology.

Contact

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Skills

Expert

RStudio, R, R shiny:

Writing functions, making packages, documenting code, creating interactive html servers.

ESRI ArcGIS Pro, QGIS:

Geospatial analyses, coordinate system management, map-making.

Creating timely deliverables and communicating effectively

Curiosity, organization, multitasking, and focus

Advanced

Python

GitHub

Big Data statistical processing, analysis, and modeling

Intermediate

SQL

Docker containerization

Google Earth Engine

ENVI

To whom it may concern,

I am writing to outline my distinct skill set, which aptly positions me as a strong candidate for this role. I am an extremely voracious learner, having earned two undergraduate and one graduate degree in five years, as well as accelerating from a Python novice to a Data Engineer in less than a year and a half. In addition to writing over ten thousand lines of code by hand, I am also adept at leveraging new tools, such as Claude, Gemini, and Copilot, to accelerate my work and place in the top 5 of 27 in our portfolio-wide AI competition. I have quickly acquired skills in Docker containerization, automating deployment using cron and GitHub Actions, hitting APIs (such as Salesforce), building ETL pipelines from start to finish, designing front-end applications using Streamlit, and using Pydantic framework for data validation.

My current role as a Data Engineer at Grassroots Carbon has been characterized by immense growth and self-driven skill acquisition, from soil carbon to coding. I tackle problems as I encounter them, often before they become business critical issues. I have ample background in geospatial datasets, having studied GIS and applied those skills to previous roles. In my current role, I have processed a dozen datasets on Uber's H3 hexagonal spatial indexing system and stored the resulting parquets on AWS S3. I have worked with COGs, geoparquet files, and geoJSONs.

In my day-to-day role, I develop modeling pipelines for soil carbon reductions and removals in python, write and rewrite code until it is clean and documented, complete numerous pull request reviews for colleagues, and read literature to enhance industry-specific knowledge. My professional goal is to always continue learning through solving complex problems that better the environment and the world.

My mantra, both in personal and professional life, is "I can do hard things." I embody this daily by meditating, cold plunging, weightlifting, and practicing other habits and routines, which I track in a personal app I built specifically for optimizing my life. I am a go-getter, a powerhouse, a never-back-down kind of person. I am confident in my abilities to learn new skills and hone those I already possess. I am always looking for interesting problems to solve, particularly those that improve the world around me. Additionally, I surround myself with smart, hardworking, high quality humans who treat others with respect and share the same goals. I truly believe I would be a great fit for this role and am willing to rise to any challenge.

Thank you for considering my application; I look forward to discussing the role further.

Best,

A handwritten signature in black ink, appearing to read 'Katy Yut', written in a cursive, flowing style.

Katherine (Katy) Yut